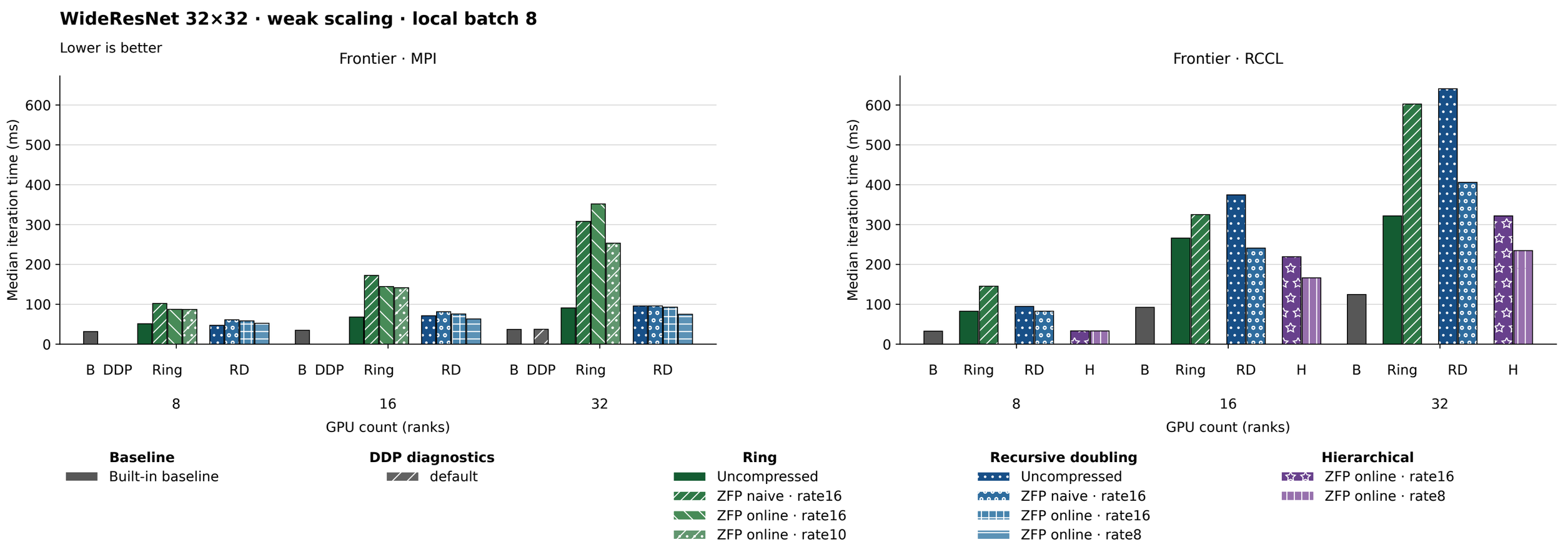

WideResNet 32×32 · weak scaling · local batch 8
Lower is better
Frontier · MPI
Frontier · RCCL
Median iteration time (ms)
0
100
200
300
400
500
600
B
DDP
Ring
RD
H
8
16
32
GPU count (ranks)
Baseline
Built-in baseline
DDP diagnostics
default
Ring
Uncompressed
ZFP naive · rate16
ZFP online · rate16
ZFP online · rate10
Recursive doubling
Uncompressed
ZFP naive · rate16
ZFP online · rate16
ZFP online · rate8
Hierarchical
ZFP online · rate16
ZFP online · rate8
Epoch 1 excluded · missing metrics are not zero · r = ZFP rate